

Name: _____ Partner: _____ Period: _____

Function Makeover Studio

A design studio applies transformation rules to “restyle” functions. The base function f is defined by the table below. Your team must construct transformed versions, match transformations to descriptions, and reverse-engineer transformations from data.

Base function f :

x	-2	-1	0	1	2
$f(x)$	4	1	0	1	4

Tier R — Remediate

Three students each applied a different transformation to f . Their tables are shown below. Match each table to the correct description by writing the letter in the blank.

Descriptions: **A.** $g(x) = f(x) - 3$ **B.** $g(x) = f(x + 2)$ **C.** $g(x) = 2f(x)$

x	-2	-1	0	1	2
Table 1: $g(x)$	1	-2	-3	-2	1

Matches description: **A**

x	-2	-1	0	1	2
Table 2: $g(x)$	8	2	0	2	8

Matches description: **C**

x	-2	-1	0	1	2
Table 3: $g(x)$	0	1	4	9	16

Matches description: **B**

Teacher Note

Table 1: $f(x) - 3$ subtracts 3 from each output ($4 - 3 = 1$, $1 - 3 = -2$, $0 - 3 = -3$, etc.).

Table 2: $2f(x)$ doubles each output ($2 \cdot 4 = 8$, $2 \cdot 1 = 2$, etc.).

Table 3: $f(x + 2)$ shifts input left 2, so $g(-2) = f(0) = 0$, $g(-1) = f(1) = 1$, $g(0) = f(2) = 4$, etc.

Tier A — Apply

Using the base function f from the scenario, construct the table of values for $g(x) = 2f(x - 1) + 3$. Show how you compute each output.

x	$x - 1$	$f(x - 1)$	$2f(x - 1)$	$g(x) = 2f(x - 1) + 3$
-1	-2	4	8	11
0	-1	1	2	5
1	0	0	0	3
2	1	1	2	5
3	2	4	8	11

State the domain and range of g based on your table.

Domain of g : $\{-1, 0, 1, 2, 3\}$ Range of g : $\{3, 5, 11\}$

Tier E — Extend

A second function h is defined by the table below. By comparing h to the base function f , identify all transformations applied to f to produce h , and write $h(x)$ in terms of f .

x	-2	-1	0	1	2
$f(x)$	4	1	0	1	4
$h(x)$	-7	-1	1	-1	-7

1. Compare outputs: how does $h(x)$ relate to $f(x)$ at each x -value?

$h(x) = -f(x) + 1$ at each x : e.g. $h(-2) = -4 + 1 = -7$, $h(-1) = -1 + 1 = 0$...wait, $h(-1) = -1$.
Check: $-f(-1) + 1 = -1 + 1 = 0 \neq -1$. **Try** $h(x) = -f(x) - 1$: $h(-2) = -4 - 1 = -5 \neq -7$. **Try**
 $a = -2$, $k = 1$: $-2(4) + 1 = -7$ ✓, $-2(1) + 1 = -1$ ✓, $-2(0) + 1 = 1$ ✓.

2. Write $h(x)$ in terms of $f(x)$. Identify a and k in $g(x) = af(x) + k$.

$h(x) = -2f(x) + 1$, so $a = -2$ and $k = 1$

3. Describe in words all transformations that map f onto h .

Vertical stretch by factor 2, reflection over the x -axis, then vertical shift up 1 unit.